

Ulrike Blume-Peytavi · Antonella Tosti
David A. Whiting · Ralph M. Trüeb (Eds.)

Hair Growth and Disorders

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With 270 Figures and 85 Tables

Professor Dr. Ulrike Blume-Peytavi

Clinical Research Center for Hair
and Skin Science (CRC)
Department of Dermatology and Allergy
Charité - Universitätsmedizin Berlin
Charitéplatz 1
10117 Berlin
Germany

Professor Dr. Antonella Tosti

Department of Dermatology
University of Bologna
Via Massarenti, 1
40138 Bologna
Italy

Professor Dr. David A. Whiting

The Hair and Skin Research
and Treatment Center
3600 Gaston Avenue
Suite 1058, Wadley Tower
Dallas, Texas 75246
USA

Professor Dr. Ralph M. Trüeb

Department of Dermatology
University Hospital of Zürich
Gloriastrasse 31
8091 Zürich
Switzerland

ISBN 978-3-540-46908-7

e-ISBN 978-3-540-46911-7

DOI 10.1007/978-3-540-46911-7

Library of Congress Control Number: 2007936859

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Cover design: Frido Steinen-Broo, eStudio Calamar, Spain

Production & Typesetting: le-tex publishing services oHG, Leipzig, Germany

Printed on acid-free paper

9 8 7 6 5 4 3 2 1

springer.com

Preface

The ideal of lifelong beautiful hair, easy to grow or to eliminate, to pigment or to depigment, to newly create or simply to replace or rearrange, is of interest to anyone interested in hair growth and its disorders, or who has patients with hair problems.

During the past 20 years, the interest in hair growth and disorders has dramatically increased. Only 20 years ago the European Hair Research Society was founded, followed by the North American Hair Research Society and the Japanese, Korean and Australasian Research Societies. The regular meetings now held on different disciplines concerning hair reflect the worldwide interest in improving our knowledge and understanding of hair and hair growth.

This textbook is testament to that interest, and contains the latest advances and scientific knowledge from the world's leading experts in hair biology and clinical trichology. In particular it covers pigmentation, genetics, hair diagnostics, pathogenesis and treatment options, hair removal and restoration techniques. Moreover, this textbook contains unique sections on tropical dermatoses of the scalp, ethnic hair and the impact of hair in forensic investigations. The book contains a chapter devoted to natural products for hair care and treatment, as well as a chapter describing hair over the ages and in art. Also, for the first time, this textbook presents a comprehensive chapter on psychocutaneous disorders of the hair and scalp.

The textbook has three sections: (i) basic aspects of human hair growth, (ii) hair and scalp disorders, and (iii)

photoepilation, surgery and hair cosmetics. Each chapter provides a synonym box, a “key features” box, a summary for the clinician, and summary tables. Graphics and clinical photographs illustrate the clinical problems and possibilities. Consequently, this textbook has a strong emphasis on visual learning.

One of the main goals in editing this textbook has been to reflect the broad spectrum of our understanding of hair growth, the clear concepts behind the pathogenesis, diagnostic procedures, and current treatment options, and to stimulate the reader to delve even deeper into the challenging task of diagnosing and helping their patients with hair problems.

We are extremely proud of, and grateful to, the world's leading experts for their excellent contribution to this textbook. The quality and diversity of topics presented are testament to the devotion of everyone involved in this project.

We would like to thank Dr. Kathrin Hillmann, who was key in assembling and preparing this book. In addition, we would like to thank the Berlin team represented by Hannelore Thomas for helping us to realize the dream of a dedicated textbook that reflects the state of the art in the “World of Hair”.

Ulrike Blume-Peytavi, Berlin, Germany

Antonella Tosti, Bologna, Italy

David Whiting, Texas, USA

Ralph Trüeb, Zürich, Switzerland

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Contributors

Hiram Larangeira de Almeida Jr

Associate Professor of Dermatology
Federal University of Pelotas
Av. Duque de Caxias 250
96.030-002 Pelotas
Brazil

F. Alan Andersen

Director, Cosmetic Ingredient Review
1101 17th Street NW, Suite 412
Washington DC
USA

Wilma F. Bergfeld

FACP Cleveland Clinic
Departments of Dermatology and Pathology
9500 Euclid Avenue, A61
Cleveland, OH 44195
USA

Ulrike Blume-Peytavi

Clinical Research Center for Hair
and Skin Science (CRC)
Department of Dermatology and Allergy
Charité – Universitätsmedizin Berlin
Charitéplatz 1
10117 Berlin
Germany

Vladimir A. Botchkarev

Professor of Cutaneous Biology
Division of Biomedical Science
School of Life Science
Richmond Road, Bradford
West Yorkshire BD71DP
UK

Valerie D. Callender

Skin and Laser Center
Suite 225
Mitchellville, MD 20721
USA

Francisco M. Camacho-Martínez

Department of Dermatology
Hospital Universitario Virgen Macarena
41009 Sevilla
Spain

Angela M. Christiano

Department of Dermatology and Genetics
& Development
Columbia University College of Physicians
and Surgeons
630 West 168th Street
New York, NY 10032
USA

Christine C. Dierickx

Director Skin and Laser Surgery Center
Beukenlaan 52
2850 Boom
Antwerp
Belgium

Zoe D. Draelos

Clinical Associate Professor
Department of Dermatology
Wake Forest University School of Medicine
Winston-Salem, North Carolina
USA

and

Dermatology Consulting Services
2444 North Main Street
High Point, NC 27262
North Carolina
USA

Marna Ericson

Department of Dermatology
University of Minnesota
MMC 98
420 Delaware St. SE
MN 55455
Minneapolis
USA

Andreas M. Finner

Trichomed- Praxis für Haarmedizin und
Haartransplantation
Klinik am Wittenbergplatz
Bayreuther Str. 36
10789 Berlin
Germany

Pia Freyschmidt-Paul

Schimmelpfengstr. 4
34613 Schwalmstadt
Germany

Natalie Garcia Bartels

Clinical Research Center for Hair
and Skin Science (CRC)
Department of Dermatology and Allergy
Charité – Universitätsmedizin Berlin
Charitéplatz 1
10117 Berlin
Germany

Uwe Gieler

Klinik für Psychosomatik und Psychotherapie
Universitätsklinikum Giessen und Marburg GmbH
Standort Giessen
Ludwigstrasse 76
35392 Giessen
Germany

Marcella Guarrera

DiSEM, Section of Dermatology
University of Genoa
Viale Benedetto XV, n.7
16132, Genova
Italy

Norbert Haas

Department of Dermatology and Allergy
Charité – Universitätsmedizin Berlin
Charitéplatz 1
10117 Berlin
Germany

Kathrin Hillmann

Clinical Research Center for Hair
and Skin Science (CRC)
Department of Dermatology and Allergy
Charité – Universitätsmedizin Berlin
Charitéplatz 1
10117 Berlin
Germany

Rolf Hoffmann

Kaiser-Joseph-Str. 262
79098 Freiburg
Germany

Maria Hordinsky

Department of Dermatology
University of Minnesota
MMC 98
420 Delaware St. SE
MN 55455
Minneapolis
USA

Nonhlanhla P. Khumalo

Division of Dermatology
Ward G23 Groote Schuur Hospital
and
The University of Cape Town
Observatory, 7925
Cape Town
South Africa

Kevin J. McElwee

Division of Dermatology
University of British Columbia
835 West 10th Avenue
Vancouver B.C.
Canada

Andrew Messenger

Department of Dermatology
Royal Hallamshire Hospital
Glossop Road
Sheffield, S10 2JF
UK

Elise A. Olsen

Division Dermatology and Oncology
Duke University Medical Center
Box 3294 DUMC
Durham, NC 27710
USA

Massimiliano Pazzaglia

Department of Dermatology
University of Bologna
Via Massarenti 1
40138 Bologna
Italy

Eva M.J. Peters

Cutaneous Neuroimmunology
Internal Medicine-Psychosomatics
Charité – Universitätsmedizin
10117 Berlin
Germany

Bianca M. Piraccini

Department of Dermatology
University of Bologna
Via Massarenti 1
40138 Bologna
Italy

Valerie A. Randall

Division of Biomedical Sciences
University of Bradford
Richmond Road, Bradford
West Yorkshire BD71DP
UK

Elizabeth K. Ross

4337 W. 14th Ave.
Vancouver, British Columbia
V6R 2X9
Canada

Jerry Shapiro

University of British Columbia Hair Research
and Treatment Center
835 West. 10th Avenue
Vancouver, British Columbia V5Z 4E8
Canada

Yutaka Shimomura

Departments of Dermatology and Genetics
& Development
Columbia University College of Physicians
and Surgeons
630 West 168th Street
New York, NY 10032
USA

Rodney Sinclair

Senior Lecturer, Department of Dermatology
University of Melbourne
St. Vincent's Hospital
PO Box 2900
Fitzroy, Victoria 3065
Australia

Mohsen Soliman

Professor of Dermatology
Faculty of Medicine Cairo University
11, El-Falaky Square
Bab Ellouk, Cairo
Egypt

Yee Jen Tai

Registrar
Department of Dermatology Services,
St. Vincent's Hospital Melbourne
PO Box 2900
Fitzroy 3065
Australia

Desmond J. Tobin

Professor of Cell Biology
Director of SLS Medical Biosciences Res. Focus Grp.
School of Life Sciences
University of Bradford
Richmond Road, Bradford
West Yorkshire BD71DP
UK

Antonella Tosti

Department of Dermatology
University of Bologna
Via Massarenti 1
40138 Bologna
Italy

Ralph M. Trüeb

Department of Dermatology
University Hospital of Zürich
Gloriastrasse 31
8091 Zürich
Switzerland

Walter P. Unger

Mount Sinai School of Medicine, New York
99 Yorkville Ave.
Suite 214
Toronto ON, M5R 3K5
Canada
and
620 Park Avenue
New York
NY 10065
USA

Robin H. Unger

620 Park Avenue
New York, NY 10021
USA

Annika Vogt

Clinical Research Center for Hair
and Skin Science (CRC)
Department of Dermatology and Allergy
Charité – Universitätsmedizin Berlin
Charitéplatz 1
10117 Berlin
Germany

David A. Whiting

The Hair and Skin Research and Treatment Center
3600 Gaston Avenue
Suite 1058, Wadley Tower
Dallas, TX 75246
USA

Abraham Zlotogorski

Department of Dermatology
Hadassah-Hebrew University Medical Center
Jerusalem 91120
Israel

Biology of the Hair Follicle

Annika Vogt, Kevin J. McElwee, Ulrike Blume-Peytavi

1

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1.1 Introduction

Over the last few decades, progress in molecular biology and genetics, as well as the development of new experimental approaches, has brought together scientists from all fields. Developmental biologists, geneticists, endocrinologists, and dermatologists now study the many diverse facets of hair follicle biology including neuroectodermal–mesenchymal interactions, immunology, pigmentation, and stem cell biology.

In humans, the main function of the hair shaft is its role as an important facet of appearance. Across cultures for centuries, the decoration and styling of scalp hair has been a means of social communication and display of social identity or status. Hair is so important in our society that hair loss, as well as the overgrowth of terminal hair on the body or face, has deleterious effects on self-esteem.

The clinical relevance of hair and hair growth impairment goes far beyond the diagnosis and treatment of hair disorders, as the following examples illustrate. More than 300 genetic conditions have hair abnormali-

ties as a component feature, and so the evaluation of hair growth is an easy, first-level diagnostic tool in pediatric dermatology and genetics. Identifying hair abnormalities can also be a useful aid in the diagnosis of suspected defects and diseases in their early stages, before more severe symptoms of progressive multi-organ involvement develop. As hair is continuously shed and allows for non-invasive collection, it can also be an important component of evidence in forensic medicine. Finally, drugs, chemicals, and biological substances accumulate and are stored in hair fiber where they can be detected and measured.

1.2 History

In the fourth century BC the philosopher Aristotle suggested that hair formed from sooty vapors exhaled through the body's pores that hardened on contact with air. These vapors were residues that the body was unable to dispose of through concoction, the process whereby food was made into nourishment (blood) for the body.

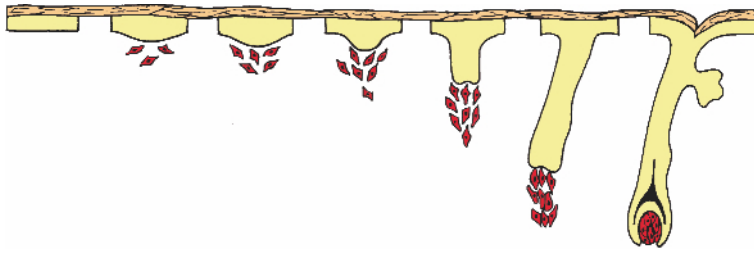


Fig. 1.1. Morphogenesis of the human hair follicle. Hair follicle formation is the result of complex sequential signaling events between the dermal mesenchyme and the overlying epithelium. Morphologically, induction, organogenesis, and cyto-differentiation phases can be determined

In the eleventh century AD, Saint and scientist Albertus Magnus, unable to determine a special role for the hair, assigned it an excretory function, and believed that scalp hair fiber was the product of gross humor and phlegm divested from the brain. Early in the sixteenth century, a microscopic study of mouse hair fiber allowed scientist William Derham to see medullae and led him to suggest the hair fibers were tubules for the evacuation of humor and perspiration from the body [18]. With the development of histological tissue analysis it became clear that hair fiber was rooted deep in the skin “implanted in the pyramidal papillae,” through which the hairs were thought to imbibe nourishment from the adjacent humors [10]. Only with the careful study of the skin and hair follicles by anatomists in the nineteenth century were hair follicle structure and its embryogenesis fully defined, and the mechanism of hair fiber growth correctly extrapolated [88].

Recent advances in molecular biology and genetics have revolutionized our understanding of hair cycling and hair growth. In particular, an increasing number of spontaneous or experimentally generated murine mutations have provided valuable insights into the functional significance of selected gene products in hair follicle morphogenesis and cycling. The identification of animal models of human diseases has helped to evaluate the relevance of candidate gene approaches.

1.3 Hair Follicle Morphogenesis

Hair follicle appendage formation involves a complex sequence of signals between the dermal mesenchyme and the overlying epithelium. The precise initiating stimulus is still to be defined. Morphologically, induction, organogenesis and cytodifferentiation phases can be determined, and a division of hair follicle formation events into eight distinct developmental stages in rodents and humans can be made [75]. The development of human hair follicles first starts between the 8th and 12th weeks of gestation. Probably in response to dermal signaling elements, which are expressed in gradients over the developing fetus, thickening of the primitive epithelium forms placodes that induce the aggregation

of underlying dermal cells to mesenchymal condensates [19] – the first visible stage in hair follicle development (Fig. 1.1). The very first hair follicle placodes are seen in the eyebrow, upper lip, and chin regions. Placode formation subsequently expands in a wave caudally and ventrally over the skin of the fetus. The dermal condensates issue instructions to the cells of their associated overlying ectodermal placodes to proliferate and initiate penetration of the dermis; stage 2 of development. As the epithelial cells grow downwards into the dermis, the dermal condensate cells lead the way. The hair follicles grow into the dermis at an angle to the skin surface, with the degree of angle determined by the location of the hair follicle [66]. The initial placode formation stage gives way to an early peg stage (stage 3) hair follicle. By 12–14 weeks’ gestation, the epithelial base of the hair pegs on the scalp invaginates to envelop the dermal cell condensates and form dermal papillae. This stage of development is described as the bulbous hair peg stage, or stage 4 in development [28].

In stage 5 at 13–16 weeks’ gestation, the superficial portions of the hair follicles subsequently develop two distinct, asymmetrical bulges of cells on the “posterior” side of the follicle, which is at an obtuse angle to the skin surface. The upper bulge closest to the skin surface eventually forms the sebaceous gland, while the lower bulge forms the location of the presumptive follicular stem cells and will later anchor the developing arrector pili muscle to the hair follicle. The arrector pili muscle itself develops independently of the hair follicle and is usually first seen in the dermis near the developing sebaceous gland [66]. The arrector pili muscle grows downwards to connect with the bulge region as the follicle pushes deeper into the dermis. Notably, the arrector pili muscle does not develop in hair follicles growing perpendicular to the skin, such as eyelash hair follicles, follicles of the external auditory canal, and those of the nasal orifice. The outer cells of the bulge, destined to become the sebaceous gland, proliferate and some differentiate into lipogenic cells that progressively accumulate lipid. Maternal hormones cause sebaceous gland hypertrophy and temporarily increase the synthesis and secretion of sebum during the second and third trimesters. With release under the influence of maternal hormones at birth, the sebaceous glands become relatively quiescent.

cent until endogenous hormone production increases in puberty. In humans, some hair follicles will develop a third superficial bulge of cells above the cells destined to become the sebaceous gland. The development of this third bulge of cells indicates the formation of an apocrine gland [30]. In humans the face and scalp are the most common locations for hair-follicle-associated apocrine gland development. While hair-follicle-associated formation of apocrine glands can be common in other mammalian species, in humans the association is relatively infrequent.

In the second trimester, the hair follicles differentiate to eventually form the seven layers of cells in concentric cylinders seen in mature hair follicles. Beginning near the bulb at the end of stage 4 or at the beginning of stage 5, a core of epithelial cells separates from the peripheral epithelial cells which later become the outer root sheath, continuous with the non-follicular epithelium. The epithelial cell core, resting on the top of the dermal papilla, further differentiates into the inner root sheath Henle, Huxley, and cuticle layers, and the central core of matrix cells that proliferate and give rise to the hair fiber cuticle, cortex and, later in terminal hairs, the medulla [9, 18]. Stage 6 is defined by the visible development and growth of the hair fiber. As the hair fiber and its inner root sheath elongate, the peripheral epithelial cells move aside to allow the cone of the central core of cells to move upwards away from the bulb. By 19–21 weeks' gestation, the developing hair follicles reach stage 7, in which the hair canals form. In stage 8 the hair follicles are fully formed and the first hair fibers erupt from the skin. The initial lanugo hair of the first anagen hair growth phase grows until 24–28 weeks of gestation.

1.4 Molecular Mediation of Hair Follicle Morphogenesis

These developmental events of hair follicle morphogenesis are controlled by a complex network of sequential activation and inactivation of autocrine, paracrine, and endocrine signaling pathways. Recent research has identified multiple regulatory factors and the essential nature of their influence on hair follicle development, but these molecular controls are not well understood [47]. How these regulators interact with each other, their relative significance, the degree of redundancy in the signaling system, and how these signals determine the development of such a complex structure, its size, and subsequent growth cycle characteristics are still largely unknown.

The source of the initial signal for hair follicle development remains unproven. Currently, evidence suggests the first signal emanates from the embryonic meso-

derm, but whether the epithelium plays a purely passive role in the induction of the first signal is not known. The earliest known molecular pathway activated during hair follicle development involves β -catenin, an intracellular mediator of gene expression. Research has shown that β -catenin is essential for keratinocyte stem cell fate decision [89]. Over-expression of stabilized β -catenin in the epidermis of transgenic mice has been shown to induce de-novo hair follicle morphogenesis, demonstrating its fundamental importance in hair follicle development [24]. In contrast, a lack of functional β -catenin results in keratinocyte differentiation to an epidermal role and a failure of hair follicle development.

Products of the WNT gene family are secreted glycoproteins that regulate cell proliferation, migration, and specification of cell fate in the embryo and adult [27]. WNT proteins are classified in part according to their ability to promote stabilization and prevent degradation of β -catenin in the cell cytoplasm. The β -catenin-dependent WNT pathway signals through cytoplasmic stabilization and accumulation of β -catenin. As it accumulates, it translocates to the cell nucleus where it forms complexes with members of the lymphoid enhancer-binding factor/T cell factor (LEF/TCF) family of DNA-binding factors to activate gene transcription. In mice, activation of WNT signaling is required in the skin for the successful induction of hair follicle development [69]. At this stage in our understanding of hair follicle embryogenesis, WNT-gene-coded proteins are the first extracellular mediators known to be involved in hair follicle development. However, it is possible that there is an even earlier gene-coded signaling mechanism that activates hair follicle development and promotes WNT gene signaling.

With the initiation of the first signaling and placode formation, a plethora of extracellular and intracellular mediators is expressed. WNT signals transduced by β -catenin and LEF-1 can elicit the expression of ectodysplasin A (EDA). Experimentally, expression of EDA, activation of its receptor (EDAR), or subsequent downstream activation of the NF- κ B pathway mediated by EDAR will each lead to hair follicle development [15]. Not surprisingly, expression of LEF-1 will also induce hair follicle formation as will expression of Noggin, transforming growth factor β 2 (TGF β 2), TGF β R-II, β 1 integrin and neural cell adhesion molecule (NCAM). Fibroblast growth factors FGF1, FGF2, FGF4, and receptor FGFR2 may also have a hair follicle inductive role, based on studies with chick embryos and the promotion of feather formation. In contrast, multiple inhibitors of hair follicle placode formation, including BMP-2, BMP-4, p75NTR and activin β a, are also expressed in embryonic skin. To add to the already crowded signaling schema, genes encoding several secreted molecules capable of inhibiting the inhibitors of hair follicle in-

duction, including Follistatin, and Gremlin, are also expressed in developing follicles. The apparently very complex interplay between activators and inhibitors of hair follicle formation, and the respective regulators of activator and inhibitor product expression likely determine the distribution of follicles in the skin [81].

With the development of hair follicles underway, yet more factors are required to regulate the hair germ in its growth and differentiation. Platelet-derived growth factor A (PDGFA) and Sonic hedgehog (Shh) are secreted by the ectodermal cells of the developing hair follicle. They are required for the condensation of the presump-

tive dermal papilla cells and for progression of the follicle placode to the peg/bulge hair stages. Asymmetric expression of Shh also polarizes the hair, ensuring outgrowth of the hair follicle in a defined direction. While activin β a can act as a placode inhibitor, activin β a in combination with Shh, hepatocyte growth factor (HGF) and its receptor (MET), transcription factor SOX18, and TGF α have been identified as driving factors during organogenesis. These and other products impel the downgrowth of the hair germ into the dermis. In stages 5–8 of hair follicle morphogenesis, the follicle differentiates into the complex structures that make up a mature hair

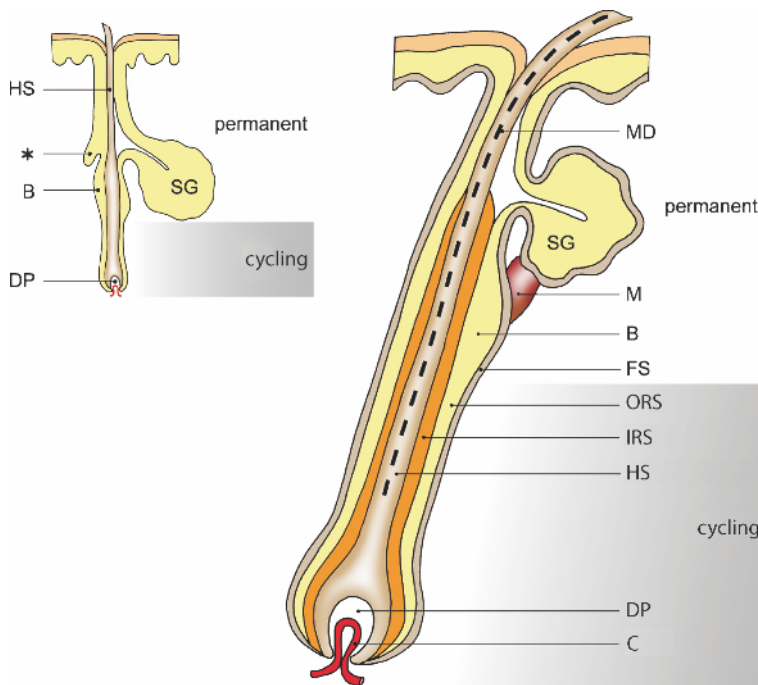


Fig. 1.2 Anatomy of the pilosebaceous unit. All hair follicles follow a common architecture. Together with the sebaceous gland (SG) and the arrector pili muscle (M), the hair follicle is part of the so-called pilosebaceous unit. The fibrous sheath (FS) and the epithelial outer and inner root sheaths (ORS, IRS) form concentric layers, which ensheath the hair shaft (HS). Hair growth results from the proliferative activity of matrix keratinocytes in the bulb, which sit on the dermal papilla (DP). The dermal papilla is a condensate of specialized mesenchymal cells with important inductive properties. It also provides nutrition via a capillary loop (C), which is especially prominent in terminal hair follicles. The permanent, superficial component has to be differentiated from the transient cycling component of the hair follicle, which includes the bulb. The morphological dividing line between these two components lies below the bulge (B) region and the insertion of the arrector pili muscle (M). The size and shape of the hair follicles, however, vary with body location and potential functions. In anagen phase, for example, vellus hair follicles from the retroauricular region (*left*) are approximately six times shorter than scalp hair terminal follicles (*right*). Each hair follicle has characteristic features. Vellus hair shafts, in contrast to terminal hair shafts, are usually devoid of a medulla (MD). Skirt-like epithelial structures (*), however, can only be found in vellus hair follicles

follicle. Yet more signaling pathways and product expression are involved in this process, including Notch1, keratinocyte growth factor (KGF), nude, bone morphogenic protein 2 (BMP2), BMP4, MOVO1, multiple HOX genes and, again, mediators such as WNT gene products and their transcriptional activators such as LEF-1. The precise interactions between all these signaling events, however, remain to be unraveled [81].

1.5 Anatomy of the Mature Pilosebaceous Unit

The hair follicle is a complex mini-organ of its own. Each hair follicle is formed by multiple mesenchymal

and epithelial cell layers, in total comprising more than 20 different cell populations. Together with the sebaceous gland and the arrector pili muscle, the hair follicle is part of the so-called pilosebaceous unit. The proportions of these components vary among the different hair follicle types (Fig. 1.2).

The general hair follicle structure can be divided into regions based on morphology, on whether the structure derives from the ectoderm or mesoderm, and on hair cycling characteristics. At its most basic, divisions can be made between the permanent, superficial structure and the transient cycling component of the hair follicle, which includes the hair bulb (Fig. 1.3). The morphological dividing line between these two components lies just below the bulge region and insertion of the arrector pili muscle [48]. The permanent portion of the hair fol-

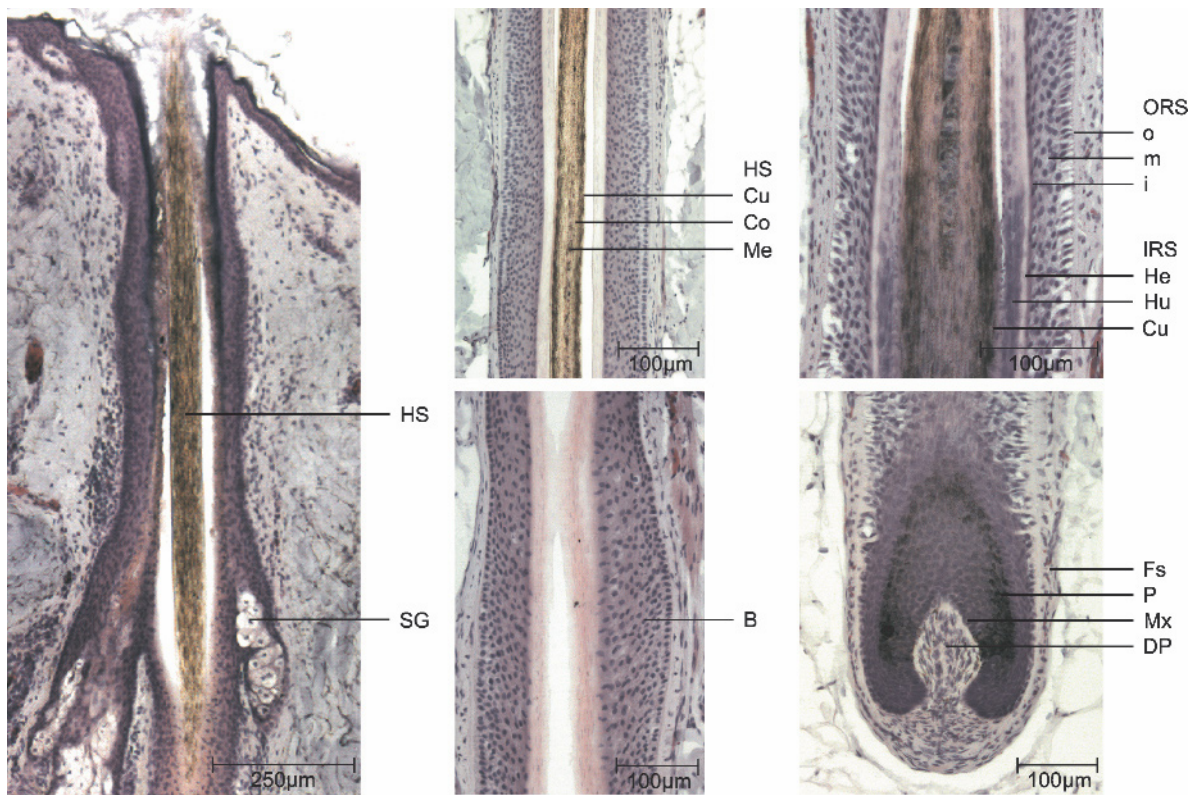


Fig. 1.3 Histomorphology of a human anagen-stage terminal hair follicle. The hair follicle itself is a complex mini-organ formed by multiple concentric mesenchymal and epithelial cell layers. The fibrous sheath (*Fs*) ensheathes the outer root sheath (*ORS*) with an outer (*o*), middle (*m*), and inner (*i*) region and the inner root sheath (*IRS*), which is further subdivided into Henle layer (*He*), Huxley layer (*Hu*), and cuticle (*Cu*). Matrix cells (*Mx*) in the hair bulb overlie the dermal papilla (*DP*) and give rise to the hair shaft (*HS*), which consists of the cuticle (*Cu*),

the cortex (*Co*) and, only in terminal hair follicles, the medulla (*Me*). Melanocytes in the bulb transfer pigment (*P*) to the hair-forming keratinocytes. The bulge region (*B*) represents a specialized compartment of the outer root sheath close to the insertion site of the arrector pili muscle, which forms a niche for epithelial and neuroectodermal stem cells as well as various immature cell populations including immature Langerhans cells and mast cells

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 licle structure can be further subdivided. From the skin surface to the point of the sebaceous gland duct opening to the hair canal it is called the infundibulum. The superficial part of the hair follicle infundibulum, the acro-infundibulum, is lined by intact epidermis including a well developed stratum corneum and a stratum granulosum. Continuous loss of epidermal differentiation occurs towards the isthmus of the lower part of the infundibulum, the infrainfundibulum.

The part between the sebaceous gland duct opening and the bulge is called the isthmus. The isthmus is a border zone relatively devoid of distinctive features though notably the thick vitreous membrane between ectoderm and mesoderm visible elsewhere in anagen-stage hair follicles becomes much thinner here. The hair follicle isthmus wall consists of two or three rows of flattened cells, whose angle of orientation can be observed to change along the length of the isthmus as the outer root sheath merges with the skin epithelium. The bulge region below the isthmus, which provides the insertion site of the arrector pili muscle, also harbors the so-called bulge region, a specialized compartment of the outer root sheath which forms a niche for epithelial and neuroectodermal stem cells as well as various immature cell populations including immature Langerhans cells and mast cells (see Sect. 1.10.1). The permanent portion of the hair follicle, the part that extends from the skin surface down to lower end of the bulge region, does not undergo significant cyclic changes. However, the presence of hair follicles, their size, and their density effectively lead to a considerable enlargement of the skin surface, which is of particular interest with regard to percutaneous penetration processes.

The inferior unit, which extends from the bulge to the base of the hair follicle bulb, can be further subdivided into the bulb and suprabulbar region. The epithelial compartments of the hair follicle include the outer root sheath and the inner root sheath, both composed of different cell sublayers, and the matrix cells (see Chap. 7). The outer root sheath extends from the matrix cells in the hair bulb up to the entry level of the sebaceous duct. Outer root sheath cells contain clear vacuolated cytoplasm filled with large amounts of glycogen. Below the isthmus, the outer root sheath is not keratinized. However, at the level of the isthmus, where the inner root sheath disintegrates, the outer root sheath keratinizes without forming granules. Outer root sheath cells express a large diversity of mediators, hormones, and receptors. The inner root sheath consists of three layers, the Henle, Huxley, and cuticle, all of which keratinize. It hardens before the presumptive hair fiber does and is, hence, thought to control the cross-sectional and longitudinal shape of the hair produced (Fig. 1.4). The mesenchymal sheath is separated from the epithe-

lial root sheaths by a vitreous or basal membrane. This whole complex is surrounded by a dense vascular network. Free nerve endings form a cuff and provide the basis for intensive piloneural interactions [49].

Hair growth results from the proliferative activity of matrix keratinocytes, which give rise to the hair shaft and the inner root sheath. They are localized in the bulb where they sit on the dermal papilla, the condensate of specialized mesenchymal cells with important inductive properties. In fact, surgical removal of the dermal papilla and the lower dermal sheath prevents hair growth, which indicates the importance of these specialized mesenchymal cells as the key signaling center in hair follicles. Nutrition of the papilla and the overlying matrix cells is provided via a capillary loop located within the dermal papilla of terminal hair follicles. Vellus hair follicle dermal papillae typically do not contain capillaries.

1.6 Hair Fiber Properties

The hair fiber, the most visible product of the hair follicle, is formed of keratin proteins, which are organized as a two-phase intracellular composite consisting of the keratin fibers embedded in an amorphous matrix. Recent genome analyses have increased the number of identified human keratins to 54, grouped into acidic type I and basic-to-neutral type II intermediate filaments (11 of the 28 type I keratins and 6 of the 26 type II keratins are hair keratins). The complex expression patterns of type I and type II hair keratins in the hair follicles as well as hair-specific expression of epithelial keratins are summarized in Table 1.1 [77]. To accommodate the growing number of keratin genes, a new consensus nomenclature for mammalian keratins was recently proposed, which also allows the incorporation of keratins from other mammalian species [76]. The matrix proteins, also referred to as keratin-associated proteins, are a major component of the hair fiber and play a crucial role in forming a strong hair shaft through a cross-linked network with keratin intermediate filaments. Generally, division occurs into high-sulfur proteins, ultra-high sulfur proteins and high-glycine-tyrosine proteins, which are differentially expressed in the cortex, cuticle, and matrix [58]. In terms of raw elements, on average, hair is composed of 50% carbon, 20% oxygen, 17% nitrogen, 6% hydrogen, and 5% sulfur. Hair also contains trace amounts of magnesium, arsenic, iron, chromium, and other metals and minerals.

The visible hair shaft of terminal hair follicles consists of three layers: cortex, cuticle, and medulla. The central medulla contains polygonal cells with a sponge-

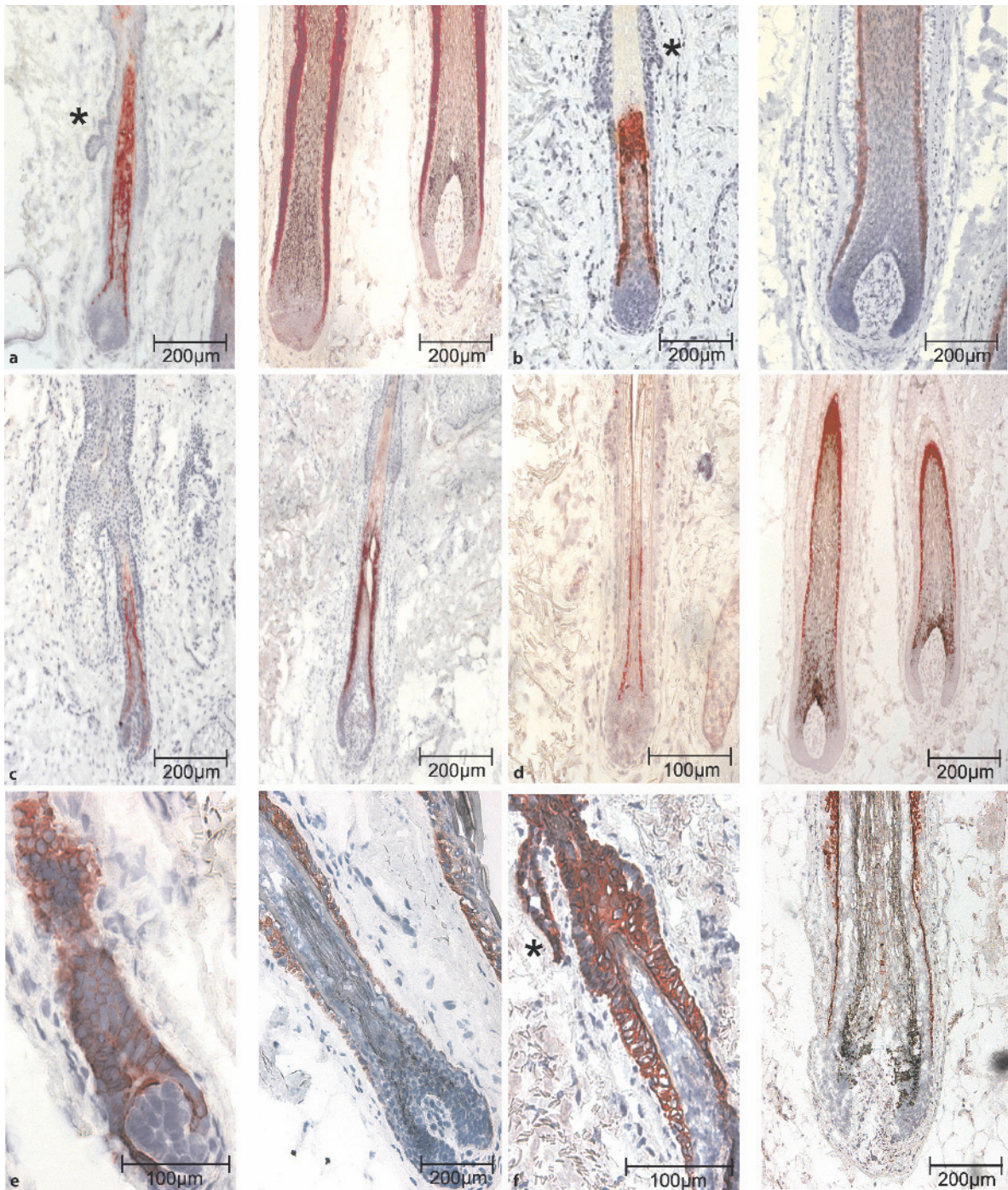


Fig. 1.4a–f Cytokeratin expression pattern in human vellus and terminal hair follicles. The expression of differentiation markers such as cytokeratins is closely related to the anatomical layers of the hair follicle. Except for some hair-follicle type-specific characteristics, such as the skirt-like epithelial structure of vellus hair follicles (*), all hair follicles follow the same architecture. This is illustrated in this figure which, in each *pan-*

el (a–f) displays one vellus hair follicles (*left*) and one terminal or intermediate hair follicle (*right*), respectively. The inner root sheath displays immunoreactivity against cytokeratin 13 (a) and trichohyalin (b), while immunoreactivity against cytokeratin 18 can predominantly be found in the Huxley layer (c). Anticytokeratin 7 specifically stains the cuticle (d). Cytokeratin 17 (e) and cytokeratin 19 (f) are markers of the outer root sheath